

Creation of a Redshift Cluster

Screenshots of the configuration of the Redshift cluster that you have created:

redshift-cluster-atm

Actions

Edit

Add partner integration

Query data

General information

Cluster identifier

redshift-cluster-atm

Status

Available

Node type

dc2.large

Endpoint

redshift-cluster-atm.ckc0a5c3fczu.us-east-1.redshift.amazonaws.com:5990/dev

Custom domain name

-

Date created

March 08, 2025, 20:48 IST

Number of nodes

2

JDBC URL

jdbc:redshift://redshift-cluster-atm.ckc0a5c3fczu.us-east-1.redshift.amazonaws.com:5990/dev

Cluster namespace ARN

arn:aws:redshift:us-east-1:640959477193:namespace:896396fa-b9a7-4160-b8e7-98fd113c3676

Multi-AZ

No

Patch version

Patch 188

ODBC URL

Driver={Amazon Redshift (x64)}; Server=redshift-cluster-atm.ckc0a5c3fczu.us-east-1.redshift.amazonaws.com; Database=dev

Namespace register status

Deregistered

Cluster configuration

Production

Storage used

0.07 of 320 GB used (0.02%)

Database configurations

Edit admin credentials

Rotate encryption keys

Edit

Database name

dev

Parameter group

Defines database parameter and query queues for all the databases.

default.redshift-2.0

Port

5990

Encryption

Enabled

AWS KMS key ID

AWS_OWNED_KMS_KEY

Audit logging

Disabled

Admin user name

awsuserurf

SSH ingestion setting (cluster public key)

ssh-rsa

AAAAB3NzaC1yc2EAAAADAQABAAQDKNx3YwSE0+NqWC1fSE3rNIZUGRXaUhd6XmP/575LtDmi2lgg75IMFmC87o13tuY41xYKvfyGnK PW17N6pAfjv4WVYbjzQevuZuiPIXZIKgNOBXUaw94+Rk61XB0DZoazcKpAihcDtJiThwkE8RxJ/Rs37Bi0M9V50LcLMOQZe2qqlohlkxgwgPmXKDx6jxRdqbGMEpsNRW0ntGSr6oaeyWEf0JeuNb38p7je5fZjMurOOD+eXCfYMW1JjCEkvCxRLUpRLxfEjAoI8DO4Tw2gumqfcXTInQjQFTQzLdKu99iff5cD3FLXHLrGoHL/bP5wqt9uNVYyaNEffkr7 Amazon-Redshift

Associated IAM roles (1)

Set default

Actions

Associate IAM roles

Create IAM role

Create, associate, or remove an IAM role. You can associate up to 50 IAM roles. You can also choose an IAM role and set it as the default for this cluster.

Find associated iam roles

IAM roles

myRedshiftRole

Status

in-sync

Role type

--

Node IP addresses (3)

Node role

Public IP address

Private IP address

Leader

98.80.39.175

172.31.38.7

Compute-0

52.55.175.46

172.31.37.60

Compute-1

13.216.185.219

172.31.36.187

Tags (0)

Manage tags

Key

Value

No tags

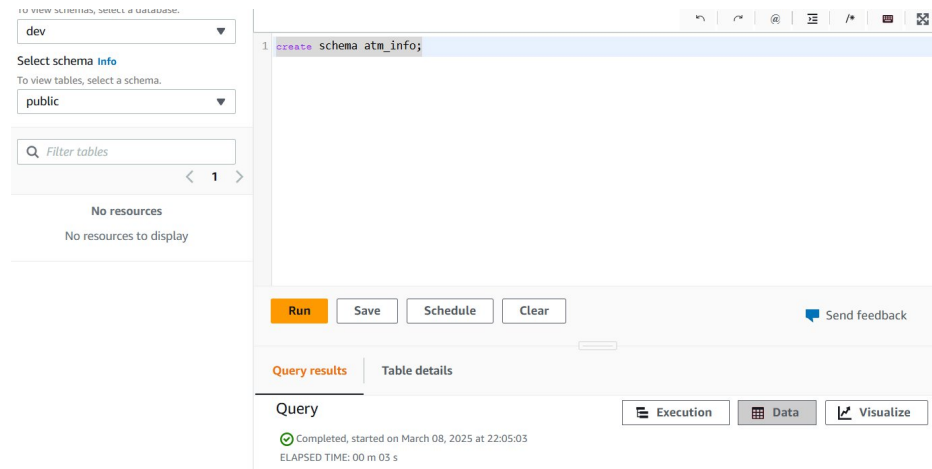
No tags are associated with this cluster.

Add tags

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Setting up a database in the Redshift cluster and running queries to create the dimension and fact tables

Schema creation



Queries to create the various dimension and fact tables with appropriate primary and foreign keys:

Creation of DIM_LOCATION table

```
create table atm_info.DIM_LOCATION(
location_id int not null DISTKEY SORTKEY,
location varchar(50),
streetname varchar(255),
street_number int,
zipcode int,
lat decimal(10,3),
lon decimal(10,3),
PRIMARY KEY(location_id)
);
```

Creation of DIM_ATM table

```
create table atm_info.DIM_ATM(
atm_id int not null DISTKEY SORTKEY,
atm_number varchar(20),
atm_manufacturer varchar(50),
atm_location_id int,
PRIMARY KEY(atm_id),
FOREIGN KEY(atm_location_id) references atm_info.DIM_LOCATION(location_id)
);
```

```
create table atm_info.DIM_ATM(
atm_id int not null DISTKEY SORTKEY,
atm_number varchar(20),
atm_manufacturer varchar(50),
atm_location_id int,
PRIMARY KEY(atm_id),
FOREIGN KEY(atm_location_id) references atm_info.DIM_LOCATION(location_id)
);
```

Run
Save
Schedule
Clear
Send feedback

Query results
Table details

Query

Execution
Data
Visualize

Completed, started on March 08, 2025 at 22:26:11
ELAPSED TIME: 06 m 45 s

Creation of DIM_DATE table

```
create table atm_info.DIM_DATE(
date_id int not null DISTKEY SORTKEY,
full_date_time timestamp,
year int,
month varchar(20),
day int,
hour int,
weekday varchar(20),
PRIMARY KEY(date_id)
);
```

```
create table atm_info.DIM_DATE(
date_id int not null DISTKEY SORTKEY,
full_date_time timestamp,
year int,
month varchar(20),
day int,
hour int,
weekday varchar(20),
PRIMARY KEY(date_id)
);
```

Run
Save
Schedule
Clear
Send feedback

Query results
Table details

Query

Execution
Data
Visualize

Completed, started on March 08, 2025 at 22:29:51
ELAPSED TIME: 00 m 26 s

Creation of DIM_CARD_TYPE table

```
create table atm_info.DIM_CARD_TYPE(  
card_type_id int not null DISTKEY SORTKEY,  
card_type varchar(30),  
PRIMARY KEY(card_type_id)  
);
```

```
create table atm_info.DIM_CARD_TYPE(  
card_type_id int not null DISTKEY SORTKEY,  
card_type varchar(30),  
PRIMARY KEY(card_type_id)  
);
```

Run

Save

Schedule

Clear

 Send feedback

Query results

Table details

Query

 Execution Data Visualize Completed, started on March 08, 2025 at 22:32:28

ELAPSED TIME: 00 m 06 s

Creation of FACT_ATM_TRANS table

```
create table atm_info.FACT_ATM_TRANS(  
trans_id bigint not null DISTKEY SORTKEY,  
atm_id int,  
weather_loc_id int,  
date_id int,  
card_type_id int,  
atm_status varchar(20),  
currency varchar(10),  
service varchar(20),  
transaction_amount int,  
message_code varchar(255),  
message_text varchar(255),  
rain_3h decimal(10,3),  
clouds_all int,  
weather_id int,  
weather_main varchar(50),  
weather_description varchar(255),
```

```
PRIMARY KEY (trans_id),
FOREIGN KEY(weather_loc_id) references atm_info.DIM_LOCATION(location_id),
FOREIGN KEY(atm_id) references atm_info.DIM_ATM(atm_id),
FOREIGN KEY(date_id) references atm_info.DIM_DATE(date_id)
);
```

Loading data into a Redshift cluster from Amazon S3 bucket

S3 bucket created and it's objects

atm-proj-ruf [Info](#)

Objects Metadata Properties Permissions Metrics Management Access Points

Objects (5) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐	Name	Type	Last modified	Size	Storage class
☐	dim_atm/	Folder	-	-	-
☐	dim_card_type/	Folder	-	-	-
☐	dim_date/	Folder	-	-	-
☐	dim_location/	Folder	-	-	-
☐	fact_atm_trans/	Folder	-	-	-

Queries to copy the data from S3 buckets to the Redshift cluster in the appropriate tables

Load data into DIM_LOCATION table

```
copy atm_info.DIM_LOCATION
from 's3://atm-proj-ruf/dim_location/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV;
```

```
copy atm_info.DIM_LOCATION
from 's3://atm-proj-ruf/dim_location/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV;
```

[Run](#) [Save](#) [Schedule](#) [Clear](#) [Send feedback](#)

[Query results](#) [Table details](#)

Query [606984](#)

[Execution](#) [Data](#) [Visualize](#)

Completed, started on March 08, 2025 at 23:01:32
ELAPSED TIME: 00 m 16 s

Load data into DIM_ATM table

```
copy atm_info.DIM_ATM
from 's3://atm-proj-ruf/dim_atm/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV
```

```
copy atm_info.DIM_ATM
from 's3://atm-proj-ruf/dim_atm/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV;
```

RunSaveScheduleClear

Send feedback

Query resultsTable details

Query [606945](#)

ExecutionDataVisualize

Completed, started on March 08, 2025 at 23:00:27

ELAPSED TIME: 01 m 31 s

Load data into DIM_DATE table

```
copy atm_info.DIM_DATE
from 's3://atm-proj-ruf/dim_date/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV timeformat 'auto';
```

```
copy atm_info.DIM_DATE
from 's3://atm-proj-ruf/dim_date/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV timeformat 'auto';
```

RunSaveScheduleClear

Send feedback

Query resultsTable details

Query [607043](#)

ExecutionDataVisualize

Completed, started on March 08, 2025 at 23:04:42

ELAPSED TIME: 00 m 20 s

Load data into DIM_CARD_TYPE table

```
copy atm_info.DIM_CARD_TYPE
from 's3://atm-proj-ruf/dim_card_type/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV;
```

```
copy atm_info.DIM_CARD_TYPE
from 's3://atm-proj-ruf/dim_card_type/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV;
```

Run Save Schedule Clear

Send feedback

Query results

Table details

Query [607075](#)

Execution Data Visualize

Completed, started on March 08, 2025 at 23:06:21
ELAPSED TIME: 00 m 15 s

Load data into FACT_ATM_TRANS table

```
copy atm_info.FACT_ATM_TRANS
from 's3://atm-proj-ruf/fact_atm_trans/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV;
```

```
copy atm_info.FACT_ATM_TRANS
from 's3://atm-proj-ruf/fact_atm_trans/'
iam_role 'arn:aws:iam::640959477193:role/myRedshiftRole'
delimiter ',' region 'us-east-1'
CSV;
```

Run Save Schedule Clear

Send feedback

Query results

Table details

Query [607104](#)

Execution Data Visualize

Completed, started on March 08, 2025 at 23:08:11
ELAPSED TIME: 00 m 32 s

